Problem 3-4

1) $P_{DC} = \frac{1}{2\pi}\int_{\omega=0^-}^{\omega=0^+} S_{xx}(\omega)\,d\omega = \int_{f=0^-}^{f=0^+} S_{xx}(f)\,df$

$= 100 \int_{0^-}^{0^+} \delta(f)\,df$

$= 100 \; V_{rms}^2$

2) $P_{total} = E[x^2(t)] = R_{xx}(0) = \int_{-\infty}^{\infty} S_{xx}(f)\,df$

$= 100 + 2 \cdot \frac{1}{2} \cdot 10 \cdot 1000$

$= 10100 \; V_{rms}^2$

3) $P_{|f|\leq 100} = \int_{-100}^{100} S_{xx}(f)\,df = 100 + 2 \cdot 9 \cdot 100 + 2 \cdot \frac{1}{2} \cdot 1 \cdot 100$

$= 2000 \; V_{rms}^2$

Problem 3-5

1) $R_{xy}(\tau) = E[X(t)Y(t+\tau)]$

$= E\left[\left[A_x\cos(\omega_0 t+\theta_x)+N_x(t)\right]\left[A_y\cos(\omega_0(t+\tau)+\theta_y)+N_y(t+\tau)\right]\right]$

$= \frac{1}{2}A_xA_y\cos(\omega_0\tau+\theta_y-\theta_x)+0+0+0$

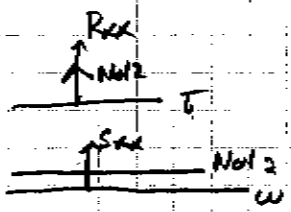
2) $R_{yx}(\tau) = \frac{1}{2}A_xA_y\cos(\omega_0\tau+\theta_x-\theta_y)+0$

S_{yx} indeholder komposant m. amplitude $\frac{1}{2}A_xA_y$ og fase $\theta_x-\theta_y$

Problem 3.6

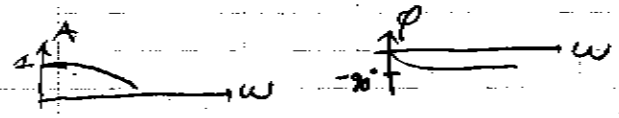
1) $R_{xx}(\tau) = \frac{N_0}{2}\delta(\tau)$ $[V_{rms}^2]$

$S_{xx}(\omega) = \frac{N_0}{2}$ $[V_{rms}^2/Hz]$



2) $H(s) = \frac{R/L}{s + R/L}$

$H(\omega) = \frac{R/L}{j\omega + R/L}$

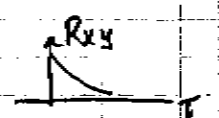


$h(t) = R/L \cdot e^{-(R/L)t}$, $t \geq 0$



3) $\mu_y = \mu_x H(0) = 0 \cdot 1 = 0$

4) $R_{xy}(\tau) = h(\tau) * R_{xx}(\tau) = \frac{N_0}{2}\frac{R}{L}e^{-\frac{R}{L}t}$, $t \geq 0$



5) enten $S_{xy}(\omega) = H(\omega)S_{xx}(\omega) = \frac{N_0}{2}\frac{R/L}{j\omega + R/L}$

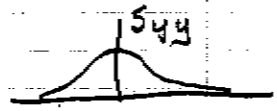
eller $S_{xy}(\omega) = \mathcal{F}\{R_{xy}(\tau)\}$

$$= \frac{N_0}{2}\frac{R}{L}\int_0^\infty e^{-R/L\,\tau}e^{-j\omega\tau}\,d\tau$$

$$= \frac{N_0}{2}\frac{R}{L}\left[\frac{e^{-(\frac{R}{L}+j\omega)\tau}}{-(\frac{R}{L}+j\omega)}\right]_0^\infty$$

$$= \frac{N_0}{2}\frac{R/L}{j\omega + R/L}$$

6) $S_{yy}(\omega) = |H(\omega)|^2 S_{xx} = \frac{N_0}{2}\frac{(R/L)^2}{\omega^2 + (R/L)^2}$



7) enten $R_{yy}(\tau) = \mathcal{F}^{-1}\{S_{yy}(\omega)\} \overset{\text{tabell}}{=} \frac{N_0}{2}\frac{R}{L}\frac{1}{2}e^{-R/L|\tau|}$



eller $\tau \geq 0$: $R_{yy}(\tau) = h(\tau) * h(-\tau) * R_{xx}(\tau)$

$$= \frac{N_0}{2}\left(\frac{R}{L}\right)^2 e^{-R/L\,\tau} * e^{R/L\,\tau}$$

$$= \frac{N_0}{2}\left(\frac{R}{L}\right)^2 \int_\tau^\infty e^{-\frac{R}{L}m}e^{-\frac{R}{L}(m-\tau)}\,dm$$

$$= \frac{N_0}{2}\left(\frac{R}{L}\right)^2 e^{R/L\,\tau}\int_\tau^\infty e^{-2\frac{R}{L}m}\,dm$$

$$= \frac{N_0}{2}\frac{R}{L}\frac{1}{2}e^{-R/L\,\tau}$$

$\tau < 0$ $R_{yy}(\tau) = R_{yy}(-\tau)$

8) enten $P_y = \sigma_y^2 = R_{yy}(0) = \frac{N_0}{2}\frac{R}{L}\cdot\frac{1}{2}$

eller $P_y = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{yy}(\omega)\,d\omega = \frac{1}{\pi}\int_0^{\infty} S_{yy}(\omega)\,d\omega$

$= \frac{1}{\pi}\frac{N_0}{2}\left(\frac{R}{L}\right)^2 \int_0^{\infty} \frac{1}{\omega^2+(R/L)^2}\,d\omega$

$= \frac{N_0}{2}\frac{R}{L}\cdot\frac{1}{2}$

9) Gauss $X(t)$ $\xrightarrow{LTI}$ Gauss $Y(t)$

$Y(t) = N(\mu_y, \sigma_y^2) = N\left(0, \frac{N_0}{2}\frac{R}{L}\frac{1}{2}\right)$

men ikke hvid (jvf. $S_{yy}(\omega)$)

10) Spm. 3 – 8 uændret

Spm. 9 fordelng af $Y(t)$ kompliceret $<$ ikke Gauss, ikke Uniform

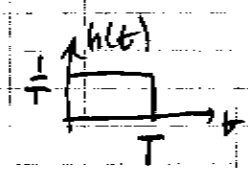
Problem 3.7

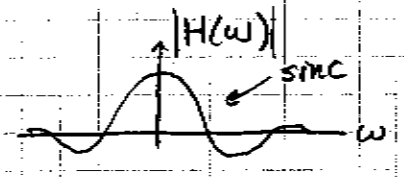
1) $R_{xx}(\tau) = \frac{N_0}{2}\delta(\tau)$ $[V_{rms}^2]$

$S_{xx}(\omega) = \frac{N_0}{2}$ $[V_{rms}^2/Hz]$

2) $H(s) = \frac{1}{T}\left(\frac{1}{s} - \frac{1}{s}e^{-sT}\right) = \frac{1}{T}\frac{1-e^{-sT}}{s}$

$H(\omega) = \frac{1}{T}\frac{1}{j\omega}(1-e^{-j\omega T}) = e^{-j\omega\frac{T}{2}}\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$





3) $Y(t) = h(t) * X(t)$

$= \int_{-\infty}^{\infty} h(m)\,X(t-m)\,dm$

$= \frac{1}{T}\int_0^T X(t-m)\,dm$

$= -\frac{1}{T}\int_t^{t-T} X(t')\,dt'$ $\quad t' = t-m$, $m = t-t'$

$= \frac{1}{T}\int_{t-T}^{t} X(t')\,dt'$ ideel midling over tid T

eller

$Y(s) = H(s)X(s) = \frac{1}{T}\left(\frac{1}{s} - \frac{1}{s}e^{-sT}\right)X(s)$

$\mathcal{L}^{-1}\downarrow$

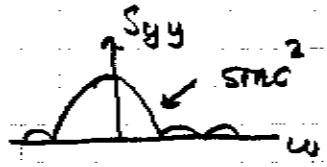
$y(t) = \frac{1}{T}\int_0^t X(t')dt' - \frac{1}{T}\int_0^{t-T} X(t')dt'$

$= \frac{1}{T}\int_{t-T}^{T} X(t')dt'$

4) $\mu_y = H(0)\cdot\mu_x = 1\cdot 0 = 0$

5) $S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$

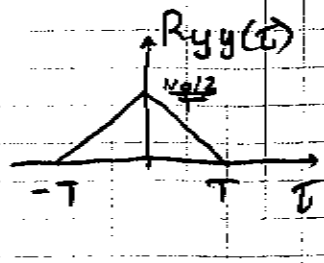
$= \left(\frac{\sin\omega T/2}{\omega T/2}\right)^2 \frac{N_0}{2}$



6) $R_{yy}(\tau) = \mathcal{F}^{-1}\{S_{yy}(\omega)\}$

$\overset{\text{tabell}}{=} \frac{N_0}{2}\frac{1}{T}\,\text{tri}\left(\frac{\tau}{T}\right)$

$= \frac{N_0}{2}\frac{1}{T}\left(1-\frac{|\tau|}{T}\right), |\tau| \le T$



7) $P_y = R_{yy}(0) = \frac{N_0}{2}\frac{1}{T}$ eller

$P_y = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{yy}(\omega)\,d\omega$

$= \frac{1}{2\pi}\cdot 2\,\frac{N_0}{2}\int_0^{\infty}\left(\frac{\sin \omega T/2}{\omega T/2}\right)^2 d\omega$

$= \frac{1}{2\pi}\,2\,\frac{N_0}{2}\,\frac{2}{T}\,\frac{\pi}{2}$

$= \frac{N_0}{2}\frac{1}{T}$

8) Gauss $X(t)$ $\xrightarrow{LTI}$ Gauss $Y(t)$

$Y(t) = N(\mu_y, \sigma_y^2) = N\left(0, \frac{N_0/2}{T}\right)$

Det ses at Var[Y] falder proportionalt med midlingsperioden T